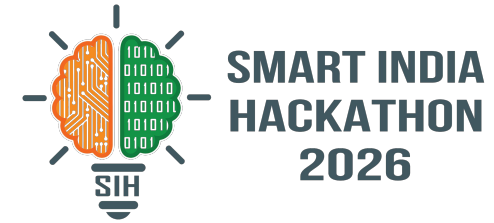
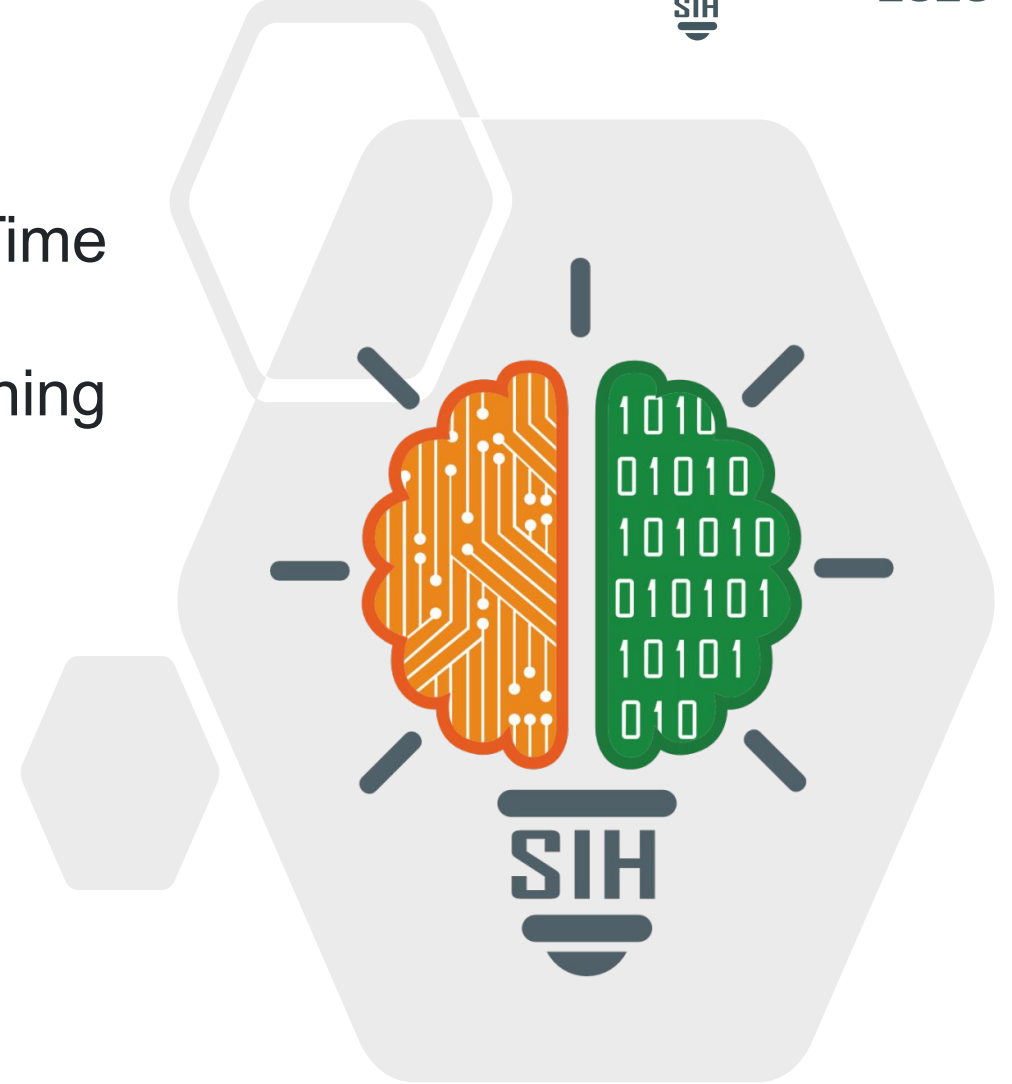


SMART INDIA HACKATHON 2026



- **Problem Statement ID** – 26104
- **Problem Statement Title-** AI-Powered Real-Time Detection and Prevention of Voice Cloning Impersonation Attacks
- **Theme-** Blockchain & Cybersecurity
- **PS Category-** Software
- **Team ID-**
- **Team Name -** Eternals



VoxShield - Real-Time AI Voice Authenticity & Fraud Prevention

◆ Proposed Solution

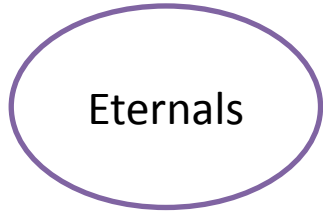
VoxShield is an **AI-powered, real-time voice authenticity verification layer** that detects AI-generated or cloned voices **during live calls**. It integrates with existing banking, telecom, and enterprise communication systems through **APIs/SDKs**, rather than replacing them.

◆ How it addresses the problem

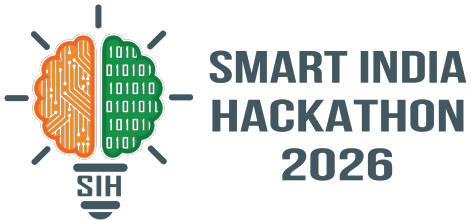
- **Real-time prevention:** Continuously analyzes live audio and raises alerts **before** fraudulent transactions or sensitive disclosures.
- **Beyond Caller ID:** Detects **acoustic, spectral and prosodic artifacts** that indicate synthetic/cloned speech.
- **Actionable risk scoring:** Combines AI confidence with **call context, transaction sensitivity and fraud history** to generate Low/Medium/High risk.
- **Cross-platform:** Designed for **bank call centers, telecom/SIP networks and enterprise collaboration platforms**.
- **Privacy-first:** Supports **edge/on-device inference** and feature-only logging instead of retaining raw voice recordings.

◆ Innovation and uniqueness

- **Streaming-first detection** — built specifically for live calls, not just uploaded recordings.
- **Context-aware AI risk fusion** — converts detection into a decision-ready security signal.
- **One Core + Multiple Adapters** — same detection engine integrates across banking, telecom and enterprise environments.
- **Indian-language & accent aware** — designed for multilingual and diverse Indian speech conditions.
- **Privacy by architecture** — edge processing and minimal audio retention are built into the design from the beginning.

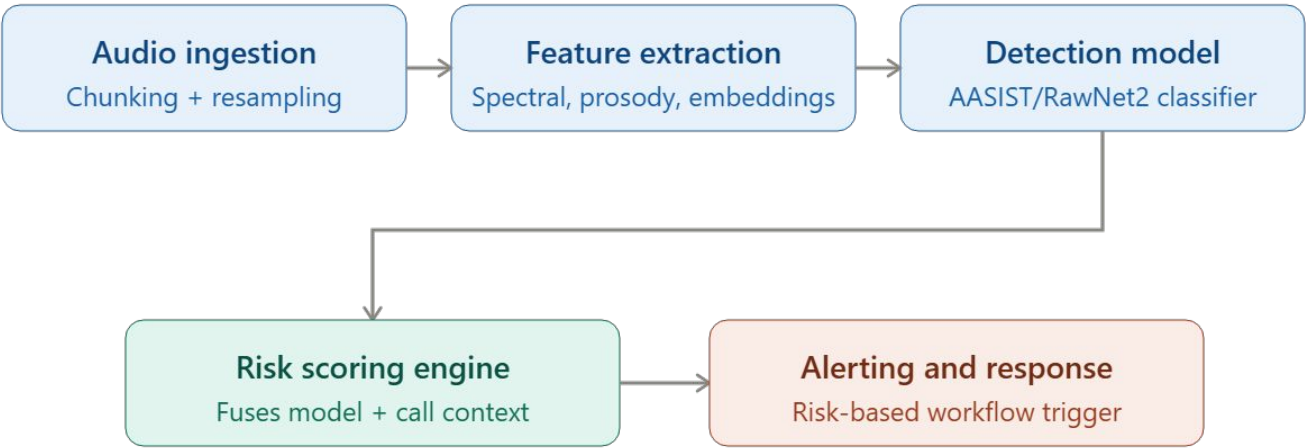


TECHNICAL APPROACH



❖ Technologies

Category	Technologies	Purpose
Programming	Python, JavaScript	ML, backend & WebRTC
AI/ML	PyTorch, AASIST / RawNet2, ECAPA-TDNN	Voice spoof detection & speaker verification
Audio Processing	Librosa, torchaudio	Feature & prosody extraction
Backend & APIs	FastAPI, gRPC, WebSocket	Real-time communication
Frontend	HTML, JavaScript, WebRTC	Live audio capture & risk UI
Infrastructure	Docker, ONNX, AWS/GCP	Deployment & edge inference
Datasets	ASVspoof, In-the-Wild, WaveFake	Model training & evaluation



1. **Audio Ingestion** → Capture live audio in **2–4 sec chunks** and resample to **16 kHz mono**.
2. **Feature Extraction** → Extract **spectral, prosodic and speaker features**.
3. **AI Detection** → AASIST/RawNet2 generates **synthetic-speech probability**.
4. **Risk Scoring** → Smooth scores and combine with **call context**.
5. **Alert & Response** → **Low: Log | Medium: Verify | High: Block/Escalate**.

❖ Feasibility

- **Technically feasible** using AASIST/RawNet2 and real-time audio processing.
- **Scalable** through API/SDK-based integration.
- **Privacy-friendly** with edge inference and minimal audio storage.
- Can be adapted for **Indian languages and accents**.

❖ Potential challenges and risks

- Background noise and poor call quality.
- Rapidly evolving voice-cloning techniques.
- Diverse Indian languages, accents and dialects.
- Real-time processing latency.
- False positives and false negatives.

❖ Strategies to Overcome

- Apply **noise reduction** and **audio preprocessing**.
- Continuously **retrain with new spoofing techniques**.
- Use **multilingual and accent-augmented datasets**.
- Optimize models using **ONNX** and **edge inference**.
- Combine AI confidence with **contextual risk scoring** and adjustable thresholds.

❖ Potential Impact

- **Banks & Financial Institutions:** Reduce voice-based financial fraud and unauthorized transactions.
- **Enterprises:** Protect sensitive approvals, credentials and confidential information.
- **Telecom Operators:** Improve security of voice communication networks.
- **Individuals:** Increase trust and safety in voice-based interactions.
- **Government Organizations:** Strengthen protection against impersonation and social-engineering attacks.

❖ Key Benefits

- **Economic:** Reduces financial losses caused by AI-powered voice fraud.
- **Security:** Enables **early detection and prevention** of voice-cloning attacks.
- **Social:** Builds trust in digital and voice-based communication.
- **Privacy:** Supports minimal audio retention and edge-based processing.
- **Scalability:** Can be integrated across banking, telecom and enterprise platforms.
- **National Impact:** Supports stronger cybersecurity resilience for India's increasingly digital communication ecosystem.

- **AASIST – Audio Anti-Spoofing using Integrated Spectro-Temporal Graph Attention Networks** — Jung et al., ICASSP 2022.
[AASIST Research Paper](#)
- **RawNet2 – End-to-End Anti-Spoofing with RawNet2** — Tak et al., ICASSP 2021.
[RawNet2 Research Paper](#)
- **ASVspoof 2019 Dataset** — Large-scale dataset containing synthesized, converted and replayed speech for anti-spoofing research.
[ASVspoof Official Website](#)
- **WaveFake Dataset** — Dataset designed for audio deepfake detection using generated speech from multiple neural architectures.
[WaveFake Paper](#)
- **Official AASIST Implementation** — PyTorch implementation with pretrained AASIST/AASIST-L models and ASVspoof evaluation setup.
[AASIST GitHub Repository](#)